

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 4 Summer 2025 (Regular)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTCVC401 - BUILDING PLANNING AND DRAWING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

1	Which principal guides for position of door & window on external wall?
	a. Aspect
	b. Prospect
	c. Form
	d. Climate
2	_____ Is the overall experience or feel of the room?
	a. Aspect
	b. Prospect
	c. Roominess
	d. Climate
3	Which rating system is used for green building in India?
	a. BREEAM
	b. LEED®
	c. GRIHA
	d. CASBEE
4	Comparative study of the human body's measurements and capabilities is called _____?
	a. Physiology
	b. Anthropometry
	c. Biology
	d. Zoology
5	Which trap is provided at the junction of house drain and street sewer?
	a. Nahani
	b. Gully trap
	c. Intercepting
	d. Q trap

6	Which method is used for lighting design in building?
	a. Lighting method
	b. Lumen method
	c. Lux method
	d. wiring method

7	What is the recommended value air change / hr. for living room?
	a. 3-6
	b. 8-12
	c. 2-4
	d. 5-7

8	Which part is used in air conditioning for dust control?
	a. exhaust fan
	b. compressor
	c. damper
	d. condenser

9	Travel distance between two staircases in residential building should be ____?
	a. 22.5
	b. 30 m
	c. 25m
	d. 15m

10	Reverberation time for Auditorium should be ____?
	a. 20 s
	b. 1 s
	c. 2 s
	d. 10 s

11	Arrangement of functional spaces as per priority and interdependency is called ____?
	a. Aspect
	b. Prospect
	c. Grouping
	d. Privacy

12	Horizontal angle of sun along the horizon is called ____?
	a. Azimuth
	b. Altitude
	c. acute angled.
	d. right angle

Q2. Solve the following.

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| A) Describe any three principals of Planning? | 6 |
| B) What are the bye laws for size, area & height of habitable rooms? | 6 |

Q3. Solve the following.

- A) Which documents are required to take building permission from local authorities ? 6
- B) Define anthropometry with its importance in planning? 6
- Q4. Solve Any Two of the following.
- A) Describe Nahani trap & gully trap with sketch? 6
- B) Explain Lumen method of lighting design with an example? 6
- C) What are the different types of wiring methods in electrification? 6
- Q5. Solve Any Two of the following.
- A) Describe working of air conditioning system with sketch ? 6
- B) Explain natural ventilation with sketch? 6
- C) What are the methods of thermal insulation used in building? 6
- Q6. Solve Any Two of the following.
- A) Explain various bye laws used in firefighting system of building? 6
- B) Define acoustics and its importance in building? 6
- C) Describe reverberation in acoustics? 6

*** End ***